

# Project Title: Global AI Tools Adoption Across Industries

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**Batch/Program:** B.Tech

**GitHub Repository:** <https://github.com/ardhigagan/Global-AI-Tools-Adoption-Across-Industries.git>

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## 1. Problem Statement

Artificial Intelligence (AI) is growing very fast, and companies all over the world are spending massive amounts of money on it. However, a major problem is that businesses do not clearly understand if these AI tools are actually helping them succeed.

Companies struggle to answer important, everyday questions: Is AI actually making us more money? Are we losing more human jobs to automation than we are creating? Does spending time training our employees to use AI actually make them work faster and better?

Without a clear way to see this data, business leaders are just guessing. They need a simple, interactive dashboard to look at the big global trends and also zoom in to see exactly how AI affects regular workers and company profits.

## 2. Solution/Features

To solve this problem, I built a complete, interactive data dashboard using Power BI. This project takes over 150,000 lines of complex data and turns it into clear, visual answers for business leaders.

The dashboard is divided into two main pages to give a complete picture:

- **The Big Picture (Macro View):** The first page shows global trends. It includes a world map showing which countries are the most digitally advanced. It also features a leaderboard showing which specific industries (like healthcare or finance) are making the most money from adopting AI, alongside total AI patents filed globally.
- **The Workforce View (Micro View):** The second page focuses on the human and technical side. It clearly shows the exact number of jobs created versus jobs lost in each industry. It also breaks down exactly which AI tools (like ChatGPT) are the most popular and what tasks companies are using them for.
- **Interactive Filters:** The entire project is connected. A user can click a button to select a specific region (like Europe) or a specific company size (like Startups), and every chart on the screen will instantly update to show data only for that specific group.

### 3. Screenshots

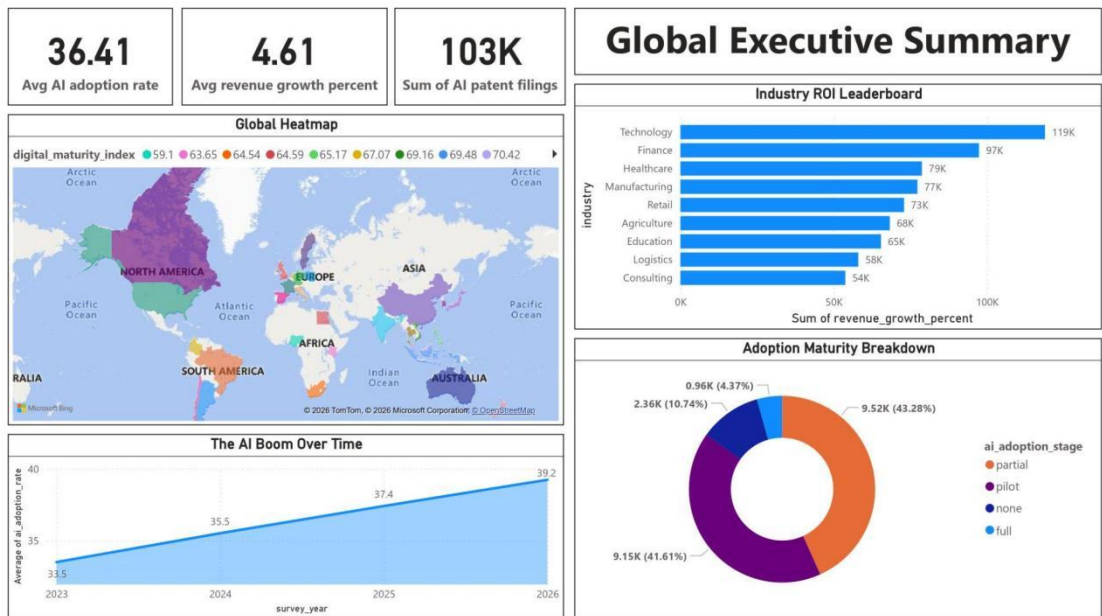


Figure 1: This view shows global AI trends, including overall adoption rates, total patents, and which industries are growing their revenue the fastest.



Figure 2: This view details the exact workforce impact, showing the balance of jobs created versus jobs lost, and maps out the most popular AI tools.

## 4. Tech Stack

To build this project from start to finish, I used a combination of powerful data analysis tools:

- **Data Cleaning (Python):** I used Python and the Pandas library inside a Jupyter Notebook. This was necessary to process the raw datasets, fix any missing values, and organize the columns so the data was clean and ready for analysis.
- **Dashboard Building (Power BI):** I imported the clean data into Power BI. I connected multiple data tables together to create a working model. I also wrote custom DAX formulas to create new calculations (like finding the true "Net Job Impact"). Finally, I designed the interactive charts and graphs.
- **Project Storage (GitHub):** I uploaded all my Python code, data files, and project documents to a public GitHub repository so the work is safely stored and easy for the evaluator to review.

## 5. Unique Points

This project stands out from standard, simple reports because of a few unique technical features:

- **The "Net Job Impact" Calculation:** Instead of just showing a scary number of how many jobs AI replaces, I created a custom formula that subtracts the jobs lost from the jobs created. This gives a much more honest and accurate view of how AI actually changes the workforce.
- **Training vs. Productivity Proof:** I built a specific scatter plot chart to test if training employees on AI is actually worth the time. By comparing "AI Training Hours" against "Productivity Change," the dashboard clearly proves whether training leads to better employee results.
- **Professional Design:** The dark, high-contrast visual theme makes the dashboard look like a modern, professional tech application, rather than a basic spreadsheet.

## 6. Future Improvements

If I had more time to expand this project in the future, I would add:

- **Predictive AI Models:** I would use Machine Learning algorithms to predict future trends. For example, the project could guess which industries will lose the most jobs next year based on current data patterns.
- **Live Data Updates:** Instead of using downloaded CSV files, I would connect the dashboard directly to live internet databases (APIs). This would allow the charts to update automatically every single day with the newest tech job postings and company funding rounds.